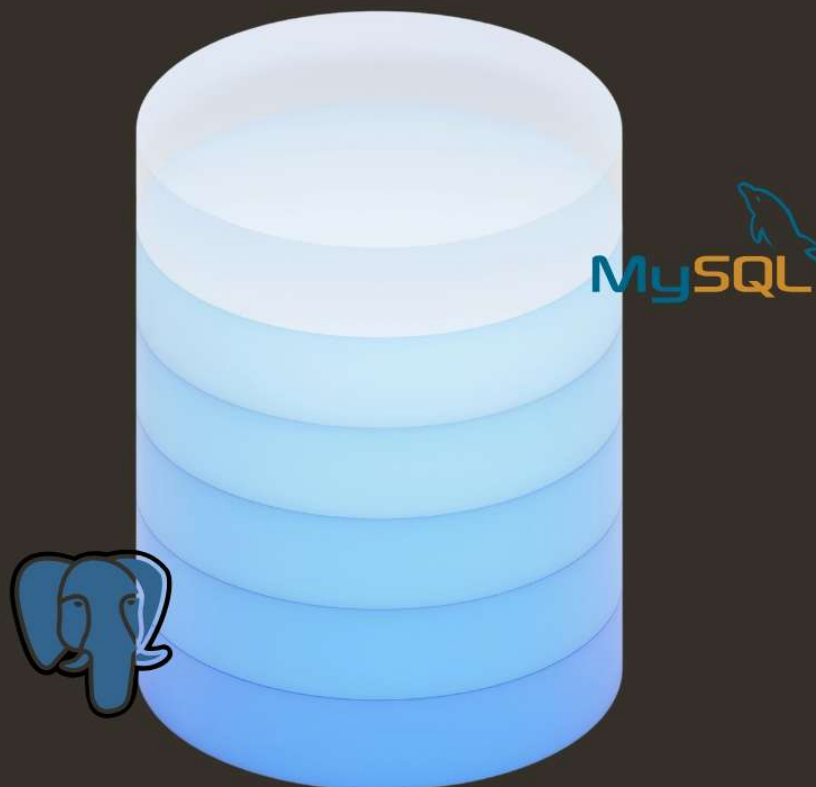


DATA QUEST

SQL for Real-World Thinking

- Aditya Varma



Preface

Welcome to Data Quest SQL

This book is written to help you learn SQL as a skill, not as a subject.

SQL is not about writing long programs. It is about understanding data and asking the right questions. It focuses on clarity over complexity. Because of this, learning SQL builds a strong foundation for working with real-world data across different domains.

The chapters in this book are arranged carefully. Each concept is introduced only when it is needed, and every idea is supported by simple queries that explain *why* something works, not just *how* to write it.

You will not just learn how to write queries, but how to think in terms of data. You will learn how information is structured, how relationships are built, and how meaningful insights are extracted.

No prior database knowledge is assumed. You are expected only to read, practice, and stay curious. This book is meant to be read slowly, like a guide. Run the queries. Modify them. Observe what changes.

Take your time. Ask questions. Make mistakes. That is how real understanding is built.

-Aditya Varma

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 **Understanding Data**

Understanding DATA!

What is Data?

Imagine waking up in the morning and checking your phone. The time you see on the screen, the number of notifications, the weather update, even the number of steps you walked yesterday all have one thing in common. They are small pieces of information describing something about the world or about you. These small pieces are what we call **data**.

Data is simply a collection of facts. It does not need to be complicated. Your name is data. Your age is data. The marks you scored in your last exam are data. Even something as simple as whether it is raining or not is also data. These are all observations, captured and recorded in some form.

On their own, these facts do not say much. The number “85” means nothing unless you know it is your exam score. The number “30” could be temperature, age, or price. Data by itself is quiet. It just exists. It waits to be understood.

When we begin to organize and interpret data, it starts becoming useful. It starts telling us something. That transformation is what makes data powerful, and that is exactly what you will learn throughout this course.

Data and Information: The Subtle Difference

Let's take a small situation. You are given a list of numbers: 85, 90, 78, 92. At first glance, these are just numbers. There is no meaning attached. This is data.

Now someone tells you, “These are the marks of a student in four subjects.” Suddenly, your understanding changes. You might think the student is doing well. You might even calculate the average. The same numbers now carry meaning.

That meaning is called **information**.

Data is raw and unprocessed. Information is what you get after understanding and interpreting that data. This distinction may seem small right now, but it is at the heart of everything you will do with SQL. You will take raw data and turn it into meaningful information.

Why Do We Store Data?

Think about how much you try to remember in a day. Names, tasks, passwords, conversations. Now imagine trying to remember the details of hundreds or thousands of things over weeks or months. It quickly becomes impossible.

This is why we store data. Consider a small grocery shop. Every day, customers come in and buy products. If the shop owner does not record anything, they will have no idea which products sell the most, which ones are rarely bought, or how much profit they are making. Everything becomes guesswork.

Now scale this up to a large company like an online shopping platform. Millions of users, millions of products, and millions of transactions happen every day. Without storing data, such a system would collapse almost instantly.

Data is stored so that it can be remembered, analyzed, and used later. It allows us to look back at what has happened and make better decisions for the future. A company can improve its services. A teacher can track student performance. A doctor can monitor patient history.

In a way, stored data acts like memory, but far more reliable and far more detailed than what a human mind can handle.

Data in the Real World

Once you start noticing it, you will realize that data is everywhere, quietly working behind the scenes. When you use a shopping app, every action is recorded. The products you search for, the items you add to your cart, the things you actually buy, all of this becomes data. This is why the app seems to “know” what you might like next.

When you watch movies on a streaming platform, it remembers what you watched, how long you watched it, and what you skipped. That data is then used to recommend new movies or shows that match your taste.

Even something as simple as booking a ride involves data. Your pickup location, destination, travel time, and fare are all stored. Over time, this data helps improve routes, estimate prices, and manage drivers more efficiently.

Social media platforms take this even further. Every like, comment, share, and scroll is recorded. Even the time you spend looking at a post becomes data. It may feel invisible, but it is constantly being generated.

Without realizing it, you are producing data every single day.

A Brief Journey Through the History of Data

Data has always been part of human life, even long before computers existed. In ancient times, people recorded information by carving symbols into stone or writing on walls. Kings and rulers kept records of taxes, land, and population using simple tools. These were early forms of data storage.

As societies grew, paper became the main method of recording information. Large registers and books were used to store records of businesses, schools, and governments. While this was an improvement, it had its own limitations. Finding specific information was slow, updating records was difficult, and storing large amounts of data required a lot of physical space.

The real transformation began with computers. Suddenly, data could be stored in digital form. It could be searched quickly, updated easily, and managed efficiently. This led to the creation of databases, which are designed specifically to organize and handle large amounts of data.

Today, we live in a world where data is generated at an enormous scale. Every click, every interaction, every transaction adds to the growing ocean of data. This modern era is often referred to as the age of data.

Some people describe data as the “new oil.” Not because it looks similar, but because of its value. Just like oil, raw data is not very useful until it is processed. But once it is refined and analyzed, it becomes incredibly powerful.

Different Forms of Data

Not all data looks neat and organized. In fact, most of it is quite messy. Some data is structured, like a table with rows and columns. This is the kind of data you might see in a spreadsheet, where everything is clearly organized. This is also the type of data you will work with most when using SQL.

Other types of data are less organized. For example, messages, images, videos, and audio files do not fit neatly into rows and columns. These are still data, but they require different ways of handling and processing. For now, it is enough to understand that data comes in many forms, but structured data is where your journey with SQL will begin.

From Data to Databases

Imagine trying to manage student records by writing them on random pieces of paper. Some names are here, some marks are there, and some details are missing altogether. It would be confusing and inefficient. Now imagine organizing all that information into a table where each student has a row, and each detail has its own column. Suddenly, everything becomes clear and easy to manage.

This organized collection of data is called a **database**.

A database allows us to store data in a structured way so that it can be easily accessed, updated, and analysed. It brings order to what would otherwise be chaos.

Why This Chapter Matters

At this point, you might feel that everything here is simple, almost obvious. And that is exactly the point. Before learning how to write SQL queries, you need to understand what you are working with. SQL is not just about writing commands. It is about asking questions to data and getting meaningful answers.

If you understand what data is, why it is stored, and how it is used, the rest of the journey becomes much easier.

Closing Thought

Right now, data may seem like nothing more than numbers, text, and records. Quiet, passive, and uninteresting.

But as you move forward, you will learn how to interact with it. You will learn how to filter it, combine it, and analyse it. And when that happens, something changes. Data stops being silent. It starts telling stories.

// Why Data Matters

If data were just stored and never used, it would be like writing a diary and never reading it again. The real value of data comes from what we *do* with it.

To understand why data matters, imagine running a small café.

Every day, customers walk in and order coffee, tea, snacks. You notice that sometimes the café is crowded, and sometimes it is empty. You might have a feeling that weekends are busier, or that people prefer cold coffee in the afternoon. But these are just guesses.

Now imagine you start recording everything. What customers ordered, at what time, on which day. After a few weeks, you look at the data and notice clear patterns. Cold coffee sells more between 2 PM and 5 PM. Weekends bring in twice as many customers. One particular snack is rarely ordered.

Suddenly, you are no longer guessing. You are making decisions based on facts. This is why data matters. It replaces assumptions with clarity.

Data and Decision Making

Every important system today runs on decisions, and most of those decisions are powered by data.

Think about a navigation app. When you enter a destination, it suggests the fastest route. It is not guessing. It is using data collected from thousands of users, traffic patterns, road conditions, and time of day.

Banks use data to decide whether to approve a loan. They look at income, past transactions, credit history, and other factors. The decision is based on patterns found in data, not just human judgment.

Even in something as simple as a classroom, a teacher can use data to track student performance. Instead of relying on memory, they can see exactly which topics students struggle with and adjust their teaching.

Data helps answer questions like:

- What is happening?
- Why is it happening?
- What should we do next?

Without data, decisions become guesses. With data, decisions become informed.

Data and Analytics

If storing data is like collecting puzzle pieces, analytics is the process of putting those pieces together to see the full picture.

Analytics is about exploring data to discover patterns, trends, and insights.

Consider a movie streaming platform. Millions of users watch content every day. The platform collects data about what people watch, how long they watch, what they skip, and what they search for.

By analysing this data, the platform can answer questions like:

- Which genres are popular right now?
- Which shows are being watched completely?
- What type of content keeps users engaged longer?

This is how recommendations work. When a platform suggests a movie “you might like,” it is not random. It is based on data analysis.

Businesses use analytics to understand customers, improve products, and increase profits. Governments use it to study population trends. Sports teams use it to analyze player performance.

Analytics turns raw data into useful knowledge.

Data and Automation

Now imagine taking it one step further. Not only are we analyzing data, but we are also using it to make systems act automatically.

This is where automation comes in.

Think about an online shopping app. When a product goes out of stock, the system can automatically detect it and notify the seller. When a user places an order, the system processes payment, updates inventory, and sends a confirmation message without any human involvement.

All of this happens because data is being continuously monitored and used to trigger actions.

Another example is email filtering. Your inbox automatically separates important emails from spam. It learns from data, patterns, and behaviour to make these decisions.

Automation saves time, reduces errors, and allows systems to operate at a scale that humans alone cannot manage.

In simple terms:

- Data tells us what is happening
- Analytics helps us understand it
- Automation helps us act on it

Types of Data

So far, we have been talking about data as if it is all the same. But in reality, data comes in different forms, and not all of it is neatly organized.

Understanding these types will help you see why SQL is designed the way it is.

- **Structured Data:**

Structured data is the most organized form of data. It is arranged in a clear format, usually in rows and columns, like a table.

Imagine a bank storing customer details. Each customer has:

- a name
- an account number
- a balance

This information fits perfectly into a table. Each row represents a customer, and each column represents a specific type of information. This kind of data is easy to store, search, and analyse. It follows a fixed structure, which makes it ideal for databases and SQL.

Whenever you see data in spreadsheets or tables, you are looking at structured data.

- **Semi-Structured Data:**

Not all data fits perfectly into tables. Sometimes, data has some organization, but not a rigid structure. This is called semi-structured data.

For example, consider a user profile stored in a flexible format. One user might have a phone number and address, while another might only have an email. The structure is not fixed, but there is still some pattern. Formats like JSON and XML are used to represent this kind of data.

It is like having notes written in a consistent style, but not in strict rows and columns. You can still understand it, but it is not as neatly organized as structured data.

- **Unstructured Data**

Now imagine data that has no predefined structure at all. This includes:

- images
- videos
- audio files
- text messages

A photo on your phone is data. A voice recording is data. A long paragraph of text is also data. But you cannot easily put these into a table with rows and columns. They require different methods to store and process.

Interestingly, most of the data in the world today is unstructured. Think about how many photos, videos, and messages are created every second.

Real-World Examples of Data Types

To make this clearer, let's connect these types to real-world systems.

In a bank, most of the data is structured. Account numbers, balances, and transaction records are all neatly organized in tables. This is why SQL is heavily used in banking systems.

On a platform like Instagram, things are more mixed. Usernames and follower counts are structured data. But photos, videos, and captions are unstructured. Comments and messages may fall somewhere in between.

Amazon uses all three types. Product listings and prices are structured. Product descriptions and reviews can be semi-structured. Images and videos of products are unstructured.

These systems combine different types of data to create a complete experience.

// What is a Database?

Imagine a school office. There are hundreds of students, each with records like name, class, marks, attendance, and contact details. Now imagine all of this information written on random sheets of paper and thrown into a drawer.

Finding one student's record would be a nightmare.

So instead, the school organizes everything neatly. Each student has a proper record. Everything is stored in a structured way so it can be easily found, updated, or reviewed.

That organized collection of data is what we call a **database**. A database is simply a system designed to **store, manage, and organize data efficiently**.

It is not just about storing data. It is about storing it in a way that makes it:

- easy to access
- easy to update
- easy to analyse

Think of a database as a highly disciplined librarian. Nothing is misplaced, everything has a position, and when you ask a question, it knows exactly where to look.

Why Not Just Use Excel?

At this point, a very natural question comes up. "If databases store tables, why not just use Excel?"

It's a fair question, because Excel *does* look similar to a database at first glance. Rows, columns, and data all sitting neatly in cells. For small tasks, Excel works perfectly fine. You can store marks of a class, track expenses, or manage a small list of items. But problems begin when things grow.

Imagine trying to manage a large online store using Excel. Thousands of products, customers, and orders. Multiple people trying to access and update the file at the same time. The file becomes slow, errors start appearing, and maintaining consistency becomes difficult.

Now imagine a bank trying to store account details in Excel. One mistake, one accidental deletion, and critical data could be lost or corrupted. There is no strong control over who can access or modify what. Databases are built to handle these problems.

They can manage huge amounts of data without slowing down. Multiple users can access the system at the same time without breaking anything. They provide security, consistency, and reliability.

Excel is like a notebook. A database is like a full library system. Both are useful, but they serve very different scales.

Database vs File System

Before databases became common, data was often stored using simple file systems.

A file system is what you use when you store files on your computer. You might have folders like “Documents,” “Photos,” or “Projects,” and inside them, files containing information. This works well for basic storage. But imagine trying to manage complex data using only files.

Suppose you are building a system for a college. You create:

- one file for students
- one file for courses
- one file for results

Now, what happens when you want to find all students enrolled in a specific course? You would have to open multiple files, search manually, and somehow connect the information. This quickly becomes messy and inefficient.

Databases solve this problem by allowing data to be connected and queried easily. Instead of jumping between files, you can ask one clear question and get a precise answer.

For example, instead of manually checking multiple files, you can simply ask:

“Show me all students enrolled in this course.”

And the database will give you the answer instantly.

Another issue with file systems is duplication. The same data might be stored in multiple files, leading to inconsistencies. If one copy is updated and another is not, errors start creeping in. Databases reduce duplication and maintain consistency by organizing data properly and linking related pieces together.

In simple terms:

- File systems store data
- Databases manage data intelligently

Types of Databases

As data grew in size and complexity, different types of databases were created to handle different needs.

- **Relational Databases (SQL):**

Relational databases are the most common and the most structured type.

They store data in tables. Each table has rows and columns. Different tables can be connected using relationships.

For example, in an online store:

- one table stores customers

- another stores orders
- another stores products

These tables are connected. A customer places orders. Orders contain products. Everything is linked. This is why they are called **relational** databases.

To work with these databases, we use SQL, which allows us to:

- retrieve data
- insert new data
- update existing data
- delete unwanted data

Relational databases are used in:

- banking systems
- e-commerce platforms
- school management systems

They are reliable, structured, and perfect for handling organized data.

- **NoSQL Databases (Basic Idea):**

As applications grew more complex, especially with social media and large-scale systems, a new type of database became popular: **NoSQL**.

Unlike relational databases, NoSQL databases do not rely strictly on tables. They are more flexible and can store data in different formats.

For example, instead of forcing all users to have the same structure, a NoSQL database can store different types of data for different users.

This is useful for applications where:

- data structure changes frequently
- large amounts of unstructured data are involved
- flexibility is more important than strict organization

NoSQL databases are commonly used in:

- social media platforms
- real-time applications
- big data systems

For now, you do not need to go deep into this. Just remember: Relational databases are structured and table-based. NoSQL databases are flexible and less structured.

Examples of Databases

Let's look at some real tools that are used in the industry.

- **MySQL:**

MySQL is one of the most widely used relational databases. It is known for being simple, fast, and reliable. Many websites and applications use MySQL to store their data.

If you have ever used a website that requires login, there is a good chance MySQL is working in the background.

- **PostgreSQL:**

PostgreSQL is another powerful relational database. It is known for being more advanced and feature-rich. It supports complex queries and is often used in systems that require high performance and reliability.

Many modern applications prefer PostgreSQL because of its flexibility and strong capabilities.

- **MongoDB:**

MongoDB is a popular NoSQL database. Instead of storing data in tables, it stores data in a flexible format similar to JSON.

This makes it easier to handle data that does not fit neatly into rows and columns, such as user profiles with varying details.

It is widely used in modern web applications, especially those dealing with large and dynamic data.